

Week 2 Regression

What is regression?

- <https://www.investopedia.com/terms/r/regression.asp>
- It is a statistical method to determine the **strength** and **character** of the **relationship** between a series of independent variables (predictors) and dependent variable (target variables)

Types of regression

- Linear Regression
- Multiple Linear Regression
- Logistic Regression
- Logarithmic Regression (and also Transformation of data)
- Polynomial (and exponential) - Univariate and Multivariate polynomial regression
- Time Series Regression
- KNN (K-Nearest Neighbour) Regression

Basic Concepts

- Predictor Variables (Independent Variables) - could be categorical, or continuous
- Target Variables (Dependent variable) - could be categorical (for classification) , or continuous (for estimation or prediction)
- **Residuals** – observed value – predicted value (this is for evaluation of model performance)

Basic Concepts – Understand the Statistics of Relationships

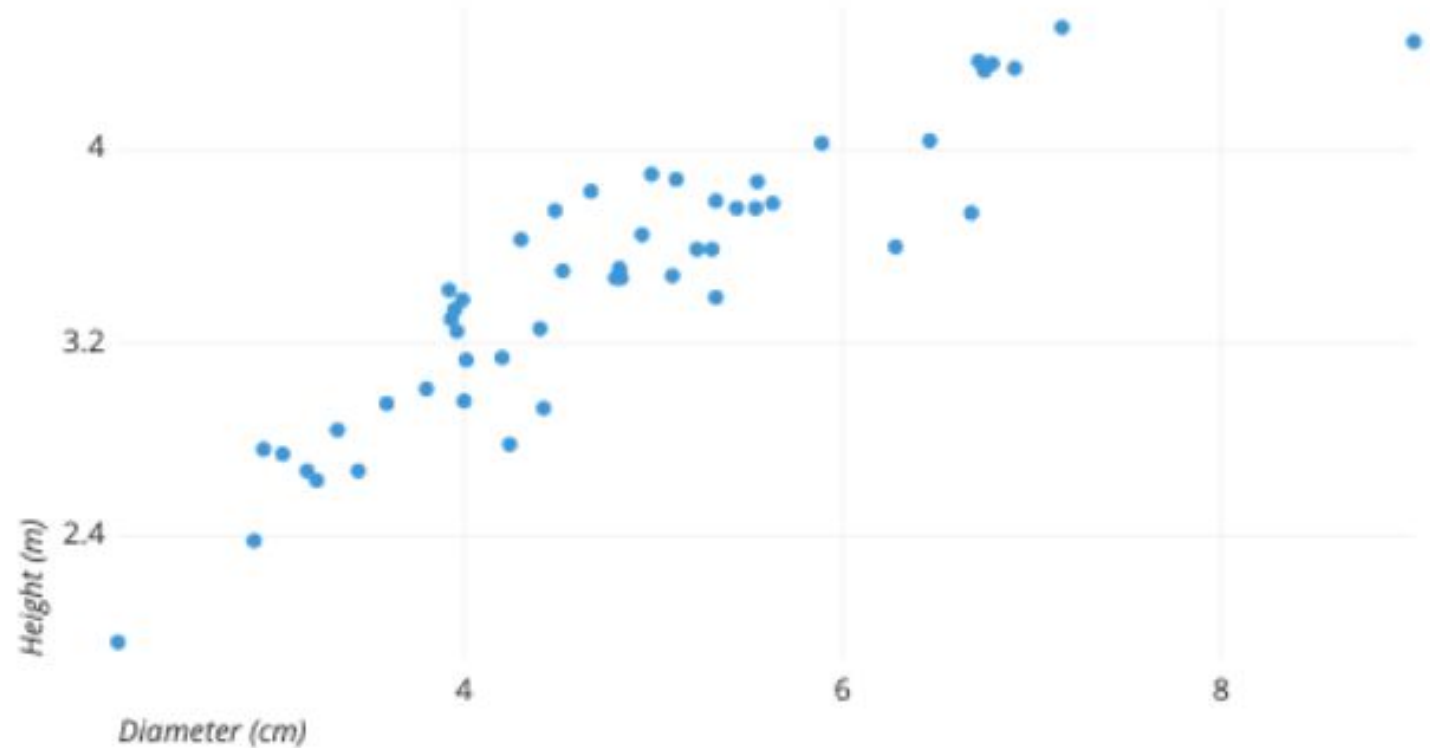
Represent relationships using Plots and Charts

Bivariate categorical data

1. If both are categorical data then use a table
2. Based on (1), find the percentage for each relevant category
3. Use Bar chart or pie chart to represent (2)

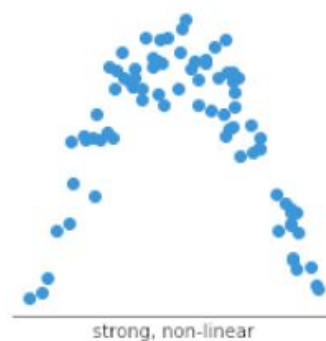
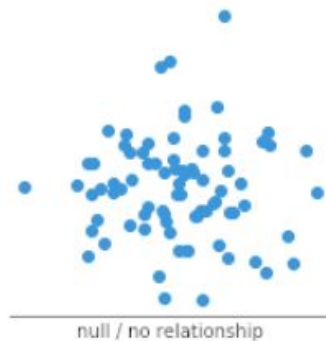
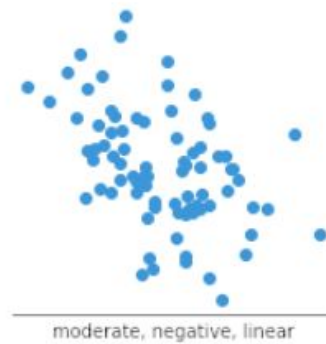
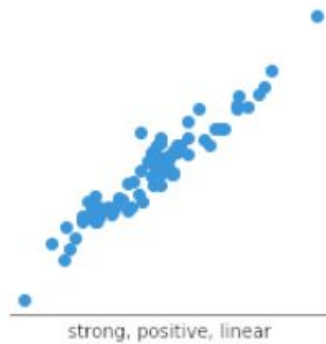
Bivariate numerical data - Scatterplot

<https://chartio.com/learn/charts/what-is-a-scatter-plot/>



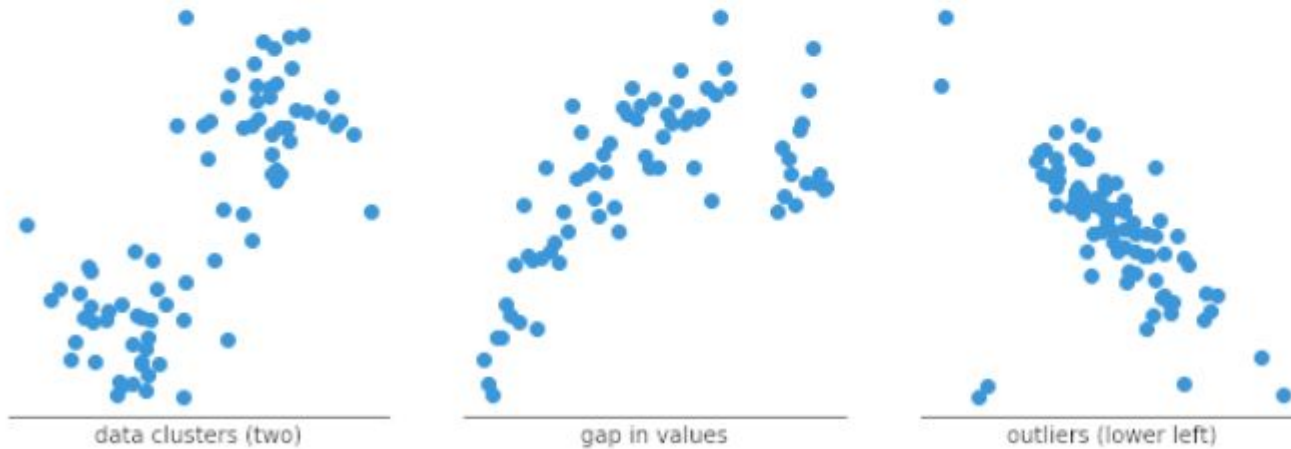
Basic Concepts – Understand the Statistics of Relationships

Identification of correlational relationships are common with scatter plots. In these cases, we want to know, if we were given a particular horizontal value, what a good prediction would be for the vertical value. You will often see the variable on the horizontal axis denoted an independent variable, and the variable on the vertical axis the dependent variable. Relationships between variables can be described in many ways: positive or negative, strong or weak, linear or nonlinear.



Basic Concepts – Understand the Statistics of Relationships

A scatter plot can also be useful for identifying other patterns in data. We can divide data points into groups based on how closely sets of points cluster together. Scatter plots can also show if there are any unexpected gaps in the data and if there are any outlier points. This can be useful if we want to segment the data into different parts, like in the development of user personas.



Basic Concepts – Quantification of Relationships

Relationship between numerical data

- Correlation coefficient (r)

Relationship between categorical data

- Association
- Hypothesis testing based on frequency contingency table (just like Chi Square)
- Example:
 - Very good video
<https://www.coursera.org/lecture/inferential-statistics/2-01-categorical-association-and-independence-2jj1Q>
 - https://www.youtube.com/watch?v=_yndRs2HEq0
 - <https://www.youtube.com/watch?v=fY8edxVRG-4>
 - https://in.sagepub.com/sites/default/files/upm-assets/67534_book_item_67534.pdf

Python Tools

- SK-Learn
- Statsmodel

How to build a simple linear regression model (for fit regression)?

- Look at excel sheet

Regression Model Evaluation

- Model Evaluation 1
- Assessing the fit of regression models
 - R-squared (and Adjusted R-squared)
 - RMSE
- Model Evaluation 2
- Assessing the classification or regression models
 - Confusion matrix
 - AUC/ROC
 - Accuracy
 - Precision
 - Recall

Model Evaluation 1

- R-squared or coefficient of determination
- SST – Sum of Squares Total (measures **variation from the mean**)
- SSE – Sum of squares Error (measures **variation of data** from model's **predicted** values)
- $R\text{-squared} = (SST - SSE) / SST$; used to explain the degree to which input (or predictor) variables explain the variation of the output (predicted or target) variables
- R-squared value: **0-1 (equivalent to 0% to 100%)- how close data is to a fitted line; R-squared value = 1.0 or 100% is perfect fit.** If R-squared is 0.9 then it indicates that 90% of the output variables is explained by the input variables
- R-squared value – **higher** indicates a better **goodness of fit** for the model
- **Increased improvement** in regression model -> increases in R-squared

- Critique of R-squared

- How to improve R-squared value?
- Add more predictors to the regression model
- The increase is artificial because the predictors are not actually improving the regression model. This is known as overfitting the model.
- If number of predictors increases -> degree of freedom increases

- Adjusted R-squared

- Adjusted R-squared - R-squared value which indicates how well the predictors fit a model, and adjusted for the number predictors in the model
- It adjusts the R-squared for predictors that are not significant in a regression model
- Suppose we add additional input variables to a model and the resultant adjusted R-squared:
 - decreases indicates that the additional input variables are not adding value to the model
 - Increases indicates that the additional input variables are adding value to the model
- Adjusted R-squared is always equal or less than R-squared
- <https://corporatefinanceinstitute.com/resources/knowledge/other/adjusted-r-squared/>

- F-Statistics

- F-test
- To answer the question “Do the independent variables reliably predict the dependent variable?”. The p-value is compared to your alpha level (typically 0.05) and, if smaller, you reject the null hypothesis (or accept the alternative hypothesis); you can conclude “Yes, the independent variables reliably predict the dependent variable”.
- Compares a model with no predictor (intercept-only model – mean of Y when $X=0$) and specified model;
- Hypotheses test
 - Null hypothesis – fit of the intercept-only model and your model are equal
 - Alternative hypothesis – fit of your model is better than the intercept-only model
- F-statistic for the overall model: example, F value = 387.18, $p < 0.0001$ (reject null hypothesis, highly significantly different)

How to Interpret Regression Results?

- From the parameter estimates, the **fitted model** is:
- Weight =
 - $-3219.68928 + 36.47545 * \text{Length}$

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	-3219.68928	346.48614	-9.29	< .0001
Length	Length (IN)	1	36.47545	1.85373	19.68	< .0001

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Additional resources interpretation of regression

- <https://stats.idre.ucla.edu/sas/output/regression-analysis/>
- <https://stats.idre.ucla.edu/sas/webbooks/reg/chapter1/regressionwith-saschapter-1-simple-and-multiple-regression/>
- <https://stats.idre.ucla.edu/sas/webbooks/reg/chapter1/regression-with-sasannotated-sas-output-for-simple-regression-analysis/>

Multiple Linear Regression Model for Prediction

- Regression model with more than 1 predictor (or input) variables
- Target variable is numeric
- $Y = a + b_1 * X_1 + b_2 * X_2 + b_3 * X_3 + \dots + b_n * X_n$ where:
 - Y is the target/predicted variable
 - a is the intercept
 - b_1, b_2, b_3, b_n are coefficients
 - $X_1, X_2, X_3, \dots, X_n$ are predictor or input variables
- Look at Example 2.2 in lab exercises
- Look at the coefficients and intercepts values

Logistic Regression

- The target variables are **categorical** variables (i.e it will be classes).
- The first example in the lab exercise is employing logistic regression as a **classifier** for three different species/classes of irises (i.e. setosa, versicolor, and virginica).
- The aim is to build a **logistic regression model**.
- We shall use iris.csv. We shall use the logistic regression model as a **prediction model** (i.e. if we use all the **4 input variables**: petal length, petal width, sepal width and sepal length, then with a given set of input values of these 4 variables, the iris class can be **predicted**).

Model Evaluation 2

- ROC/AUC

- A Receiver Operating Characteristic (**ROC**) curve
 - Evaluate the performance of a **classification model**
 - Values are based on the **confusion matrix**
 - A graph with the **x-axis** values as the **False Positive Rate** (FPR) and the **y-axis** values as the **True Positive Rate** (TPR).
- The Area Under the Curve (**AUC**) for the ROC
 - $AUC < 0.5$ then it is a bad model

- Basic Concepts

- <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>

Basic Concepts

- True Positive Rate (TPR)

- Known as **Recall**
- **TP** – True Positive
- **FN** – False Negative

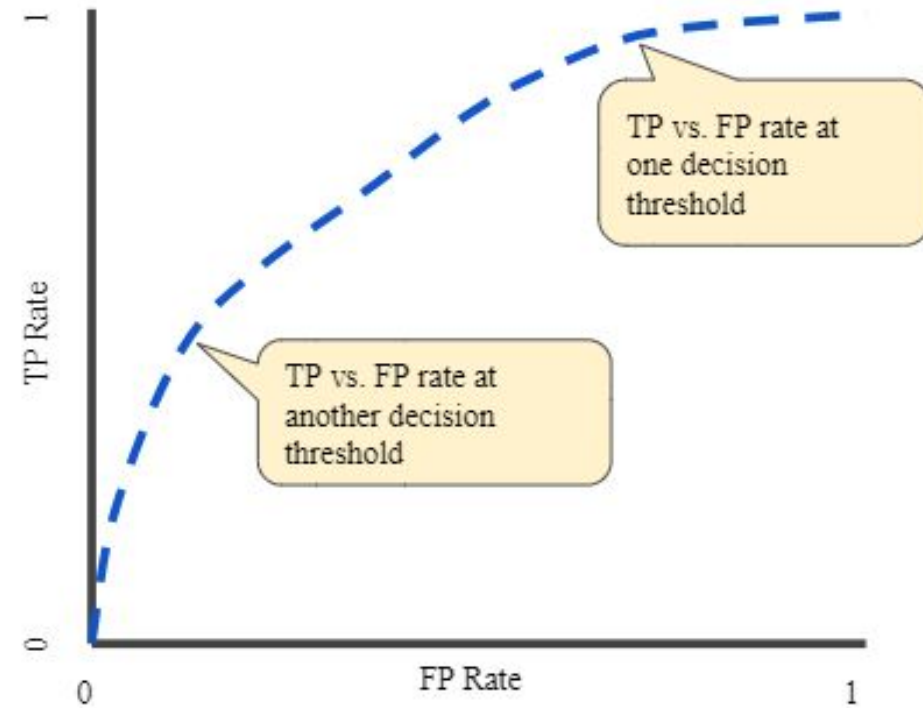
$$TPR = \frac{TP}{TP + FN}$$

- False Positive Rate (FPR)

- **FP** – False Positive
- **TN** – True Negative

$$FPR = \frac{FP}{FP + TN}$$

ROC curve: TPR vs FPR

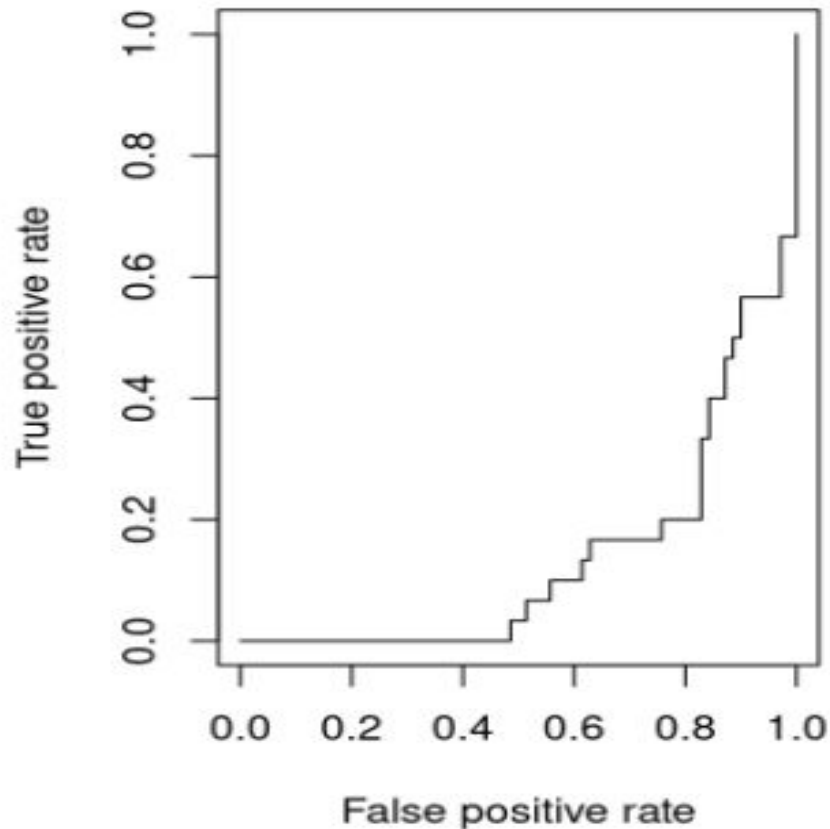


Basic Concepts

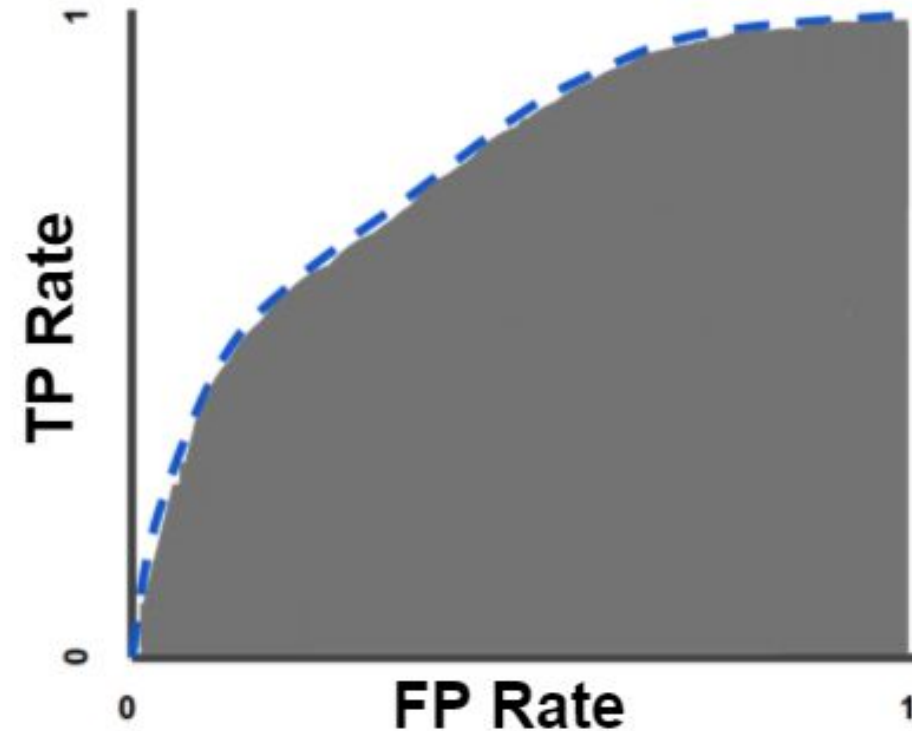
- AUC – Area under ROC Curve
 - Integral calculus from (0,0) to (1,1)
 - $AUC < 0.5$ – bad classifier model
 - $AUC > 0.5$ – good classifier model

AUC – bad classifier model

Curve (AUC: 0.14)



AUC – good classifier model



Model Evaluation Metrics: Accuracy, Precision and Recall

Accuracy Score - Accuracy It is the ratio of number of correct predictions to the total number of input samples.

Note: In the formula below, **FPR is R**

https://scikit-learn.org/stable/auto_examples/model_selection/plot_precision_recall.html

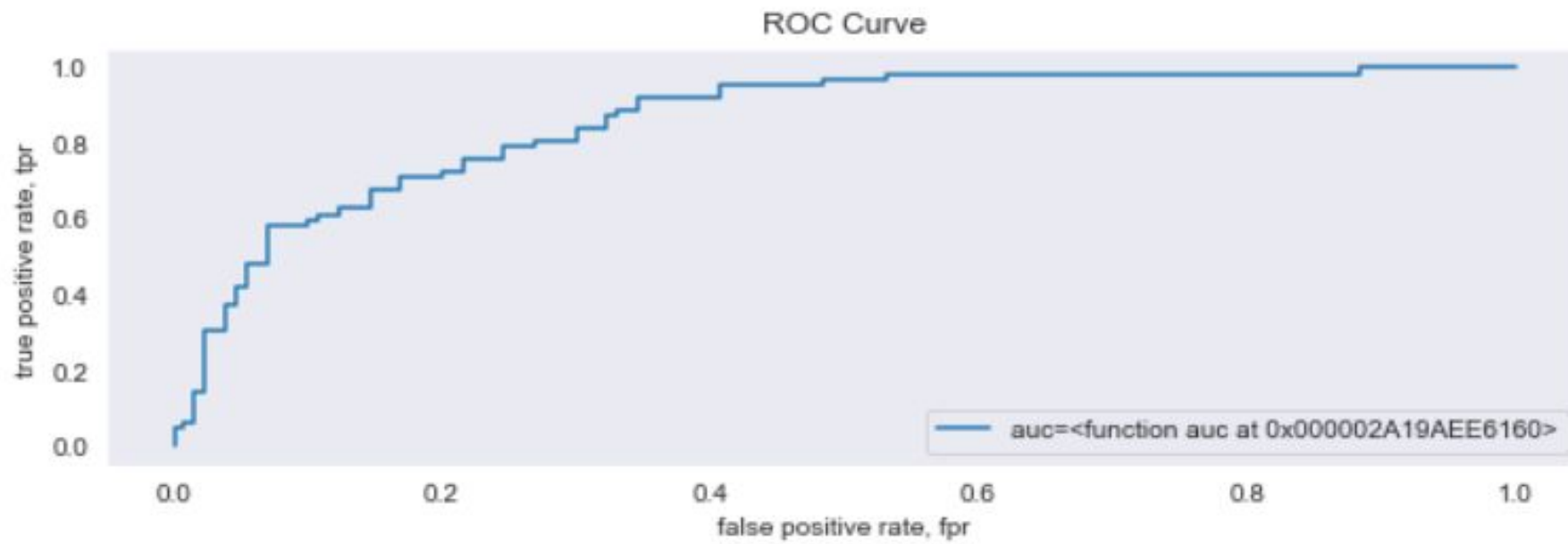
Precision (P) is defined as the number of true positives (T_p) over the number of true positives plus the number of false positives (F_p).

$$P = \frac{T_p}{T_p + F_p}$$

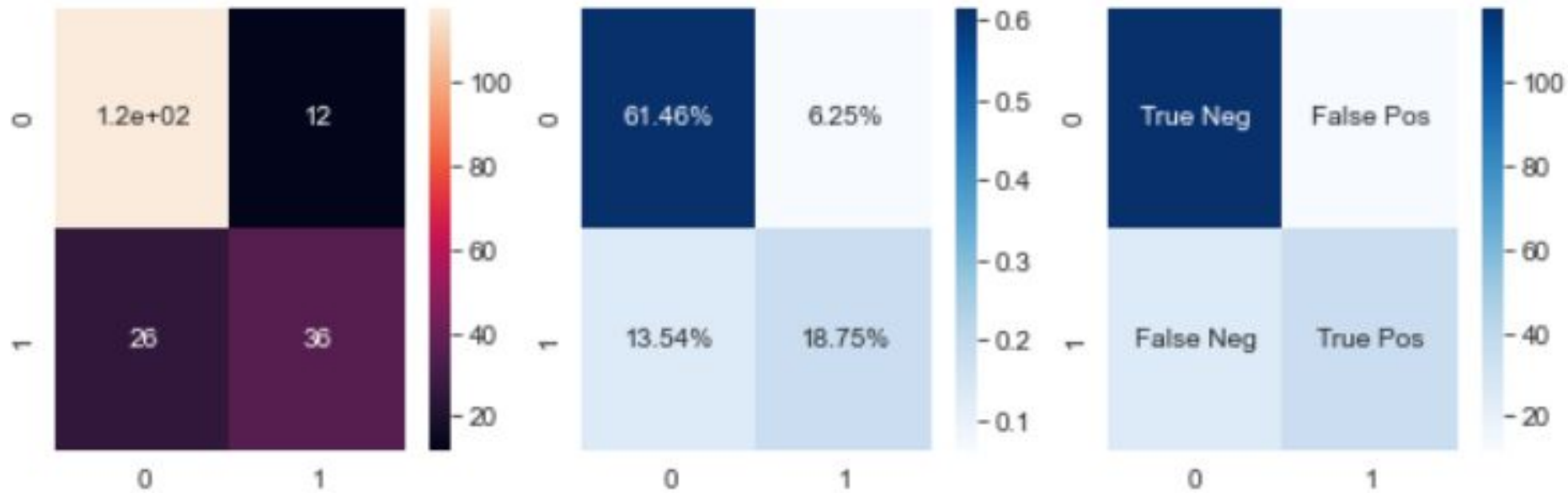
Recall (R) is defined as the number of true positives (T_p) over the number of true positives plus the number of false negatives (F_n).

$$R = \frac{T_p}{T_p + F_n}$$

- Confusion Matrix
- See next slide



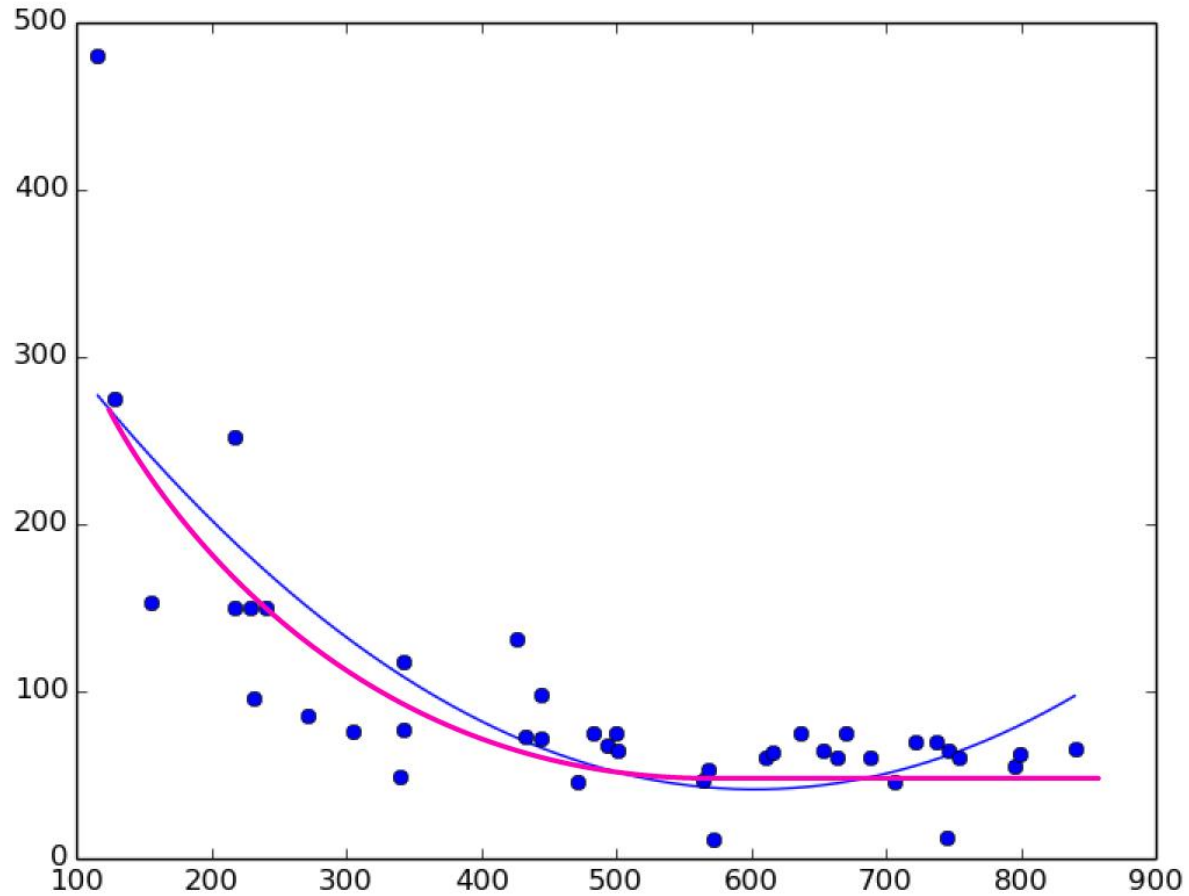
ROC Curve and
Confusion Matrix



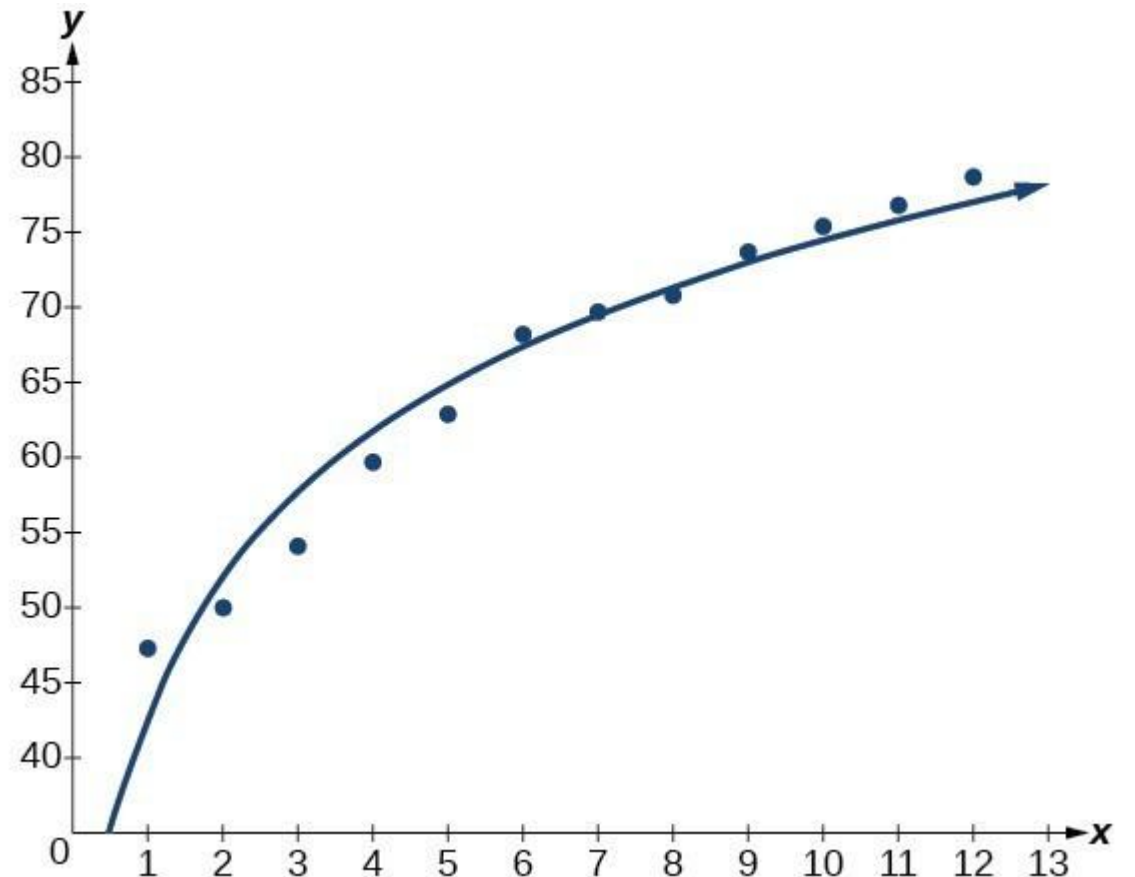
Logarithm Regression Model

- Logarithm Regression Model
 - Equation is $Y = a + b \cdot \ln(X)$
 - X must be greater than 0
 - a is the intercept
 - b is coefficient
 - When $b > 0$, model is increasing
 - When $b < 0$, model is decreasing
 - ln – natural log
 - used to model situations where **growth or decay accelerates rapidly** at first and then **slows over time**.
 - <https://courses.lumenlearning.com/ivytech-collegealgebra/chapter/build-a-logarithmic-model-from-data/>

Examples of Logarithm Regression Models



$b < 0$

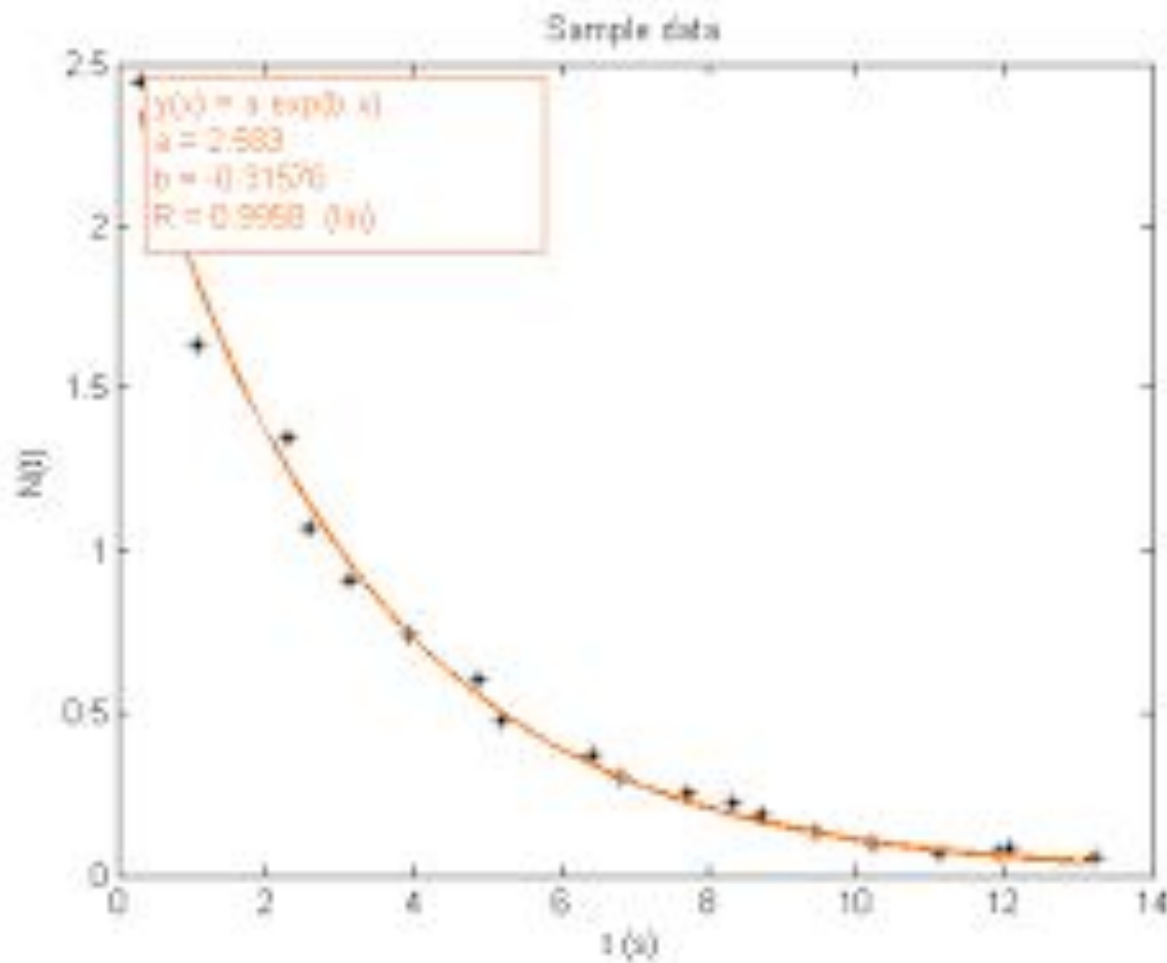


$b > 0$

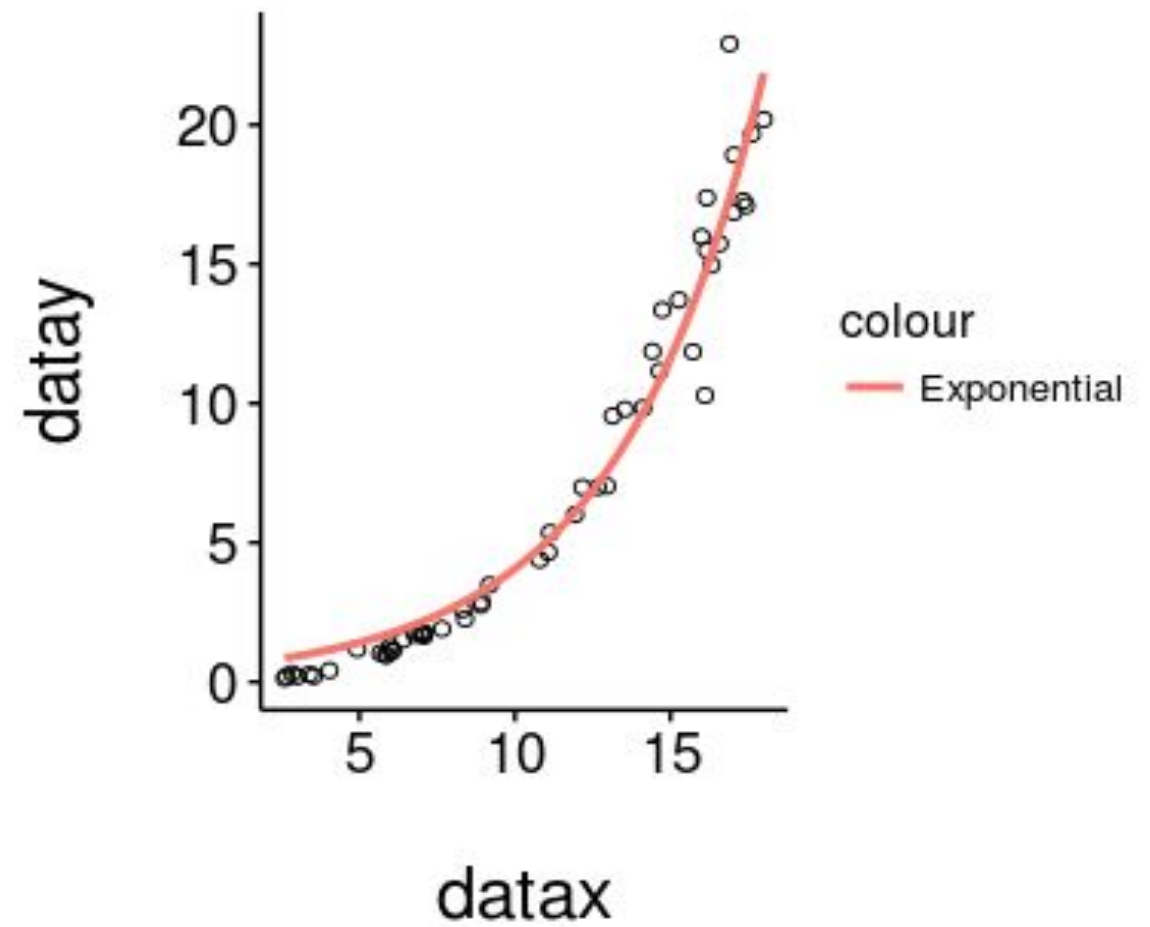
Exponential Regression Model

- Exponential Regression Model
 - Equation is $Y = ab^x$
 - a is **not equal** to 0
- Exponential regression model is used to **model situations** in which:
 - growth **begins slowly** and then **accelerates rapidly** without bound (to infinity),
 - or where **decay begins rapidly** and **then slows down to get closer and closer** to zero.

Examples of Exponential Regression Models



$b < 0$



$b > 0$

Transformation of Data

- Logarithm Transformation
- Exponential Transformation
- Logarithmic Transformation of data helps reduce skewness in the data. You can could conduct **log transformation** for continuous data.
- Look at Example 4.1 in Lab Exercises

Polynomial Regression

- Polynomial regression is a **special case of linear regression where we fit a polynomial equation** on the data with a **curvilinear relationship** between the target variable and the independent variables (look at **the scatterplot**).
- Univariate Polynomial Regression - Look at Example 5.1 in lab exercises
- Multivariate Polynomial Regression – Look at Example 5.2 in lab exercises
- Univariate - Degree 2 : $Y = \text{intercept} + aX_1 + bX_1^2 + \text{residual (or bias)}$
 - a, b are weights
 - X_1 is a predictor
 - 2 - degree
- Bivariate – Degree 2 : $Y = \text{intercept} + aX_1 + bX_2 + cX_1^2 + dX_2^2 + (eX_1X_2 - \text{if there is interaction}) + \text{residual (or bias)}$
 - $a, b, c,$ and d are weights
 - X_1 and X_2 are predictors
 - 2 - degree

Polynomial Regression

- Multivariate – Degree n and single predictor

- $Y = \theta_0 + \theta_1 X + \theta_2 X^2 + \theta_3 X^3 + \dots + \theta_n X^n$
- θ_0 is bias
- θ_n are weights
- X is a predictor
- n – degree

- Multivariate – Degree n and m number of predictors

$$\begin{aligned} Y = & \theta_0 + \theta_{1,1} X_1 + \theta_{1,2} X_2 + \theta_{1,3} X_3 + \dots + \theta_{1,m} X_m + \\ & \theta_{2,1} X_1^2 + \theta_{2,2} X_2^2 + \theta_{2,3} X_3^2 + \dots + \theta_{2,m} X_m^2 + \\ & \theta_{3,1} X_1^3 + \theta_{3,2} X_2^3 + \theta_{3,3} X_3^3 + \dots + \theta_{3,m} X_m^3 + \\ & \cdot \\ & \cdot \\ & \cdot \\ & \theta_{n,1} X_1^n + \theta_{n,2} X_2^n + \theta_{n,3} X_3^n + \dots + \theta_{n,m} X_m^n \end{aligned}$$

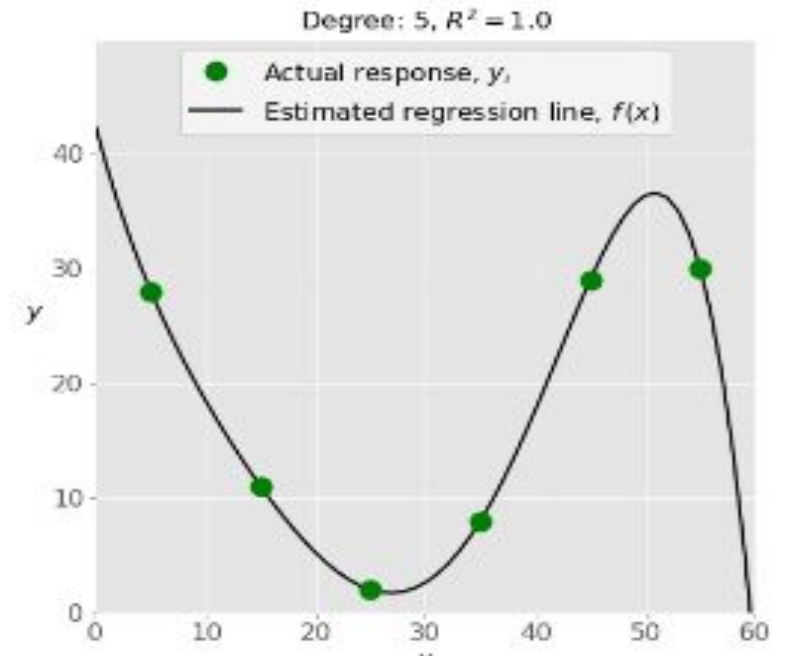
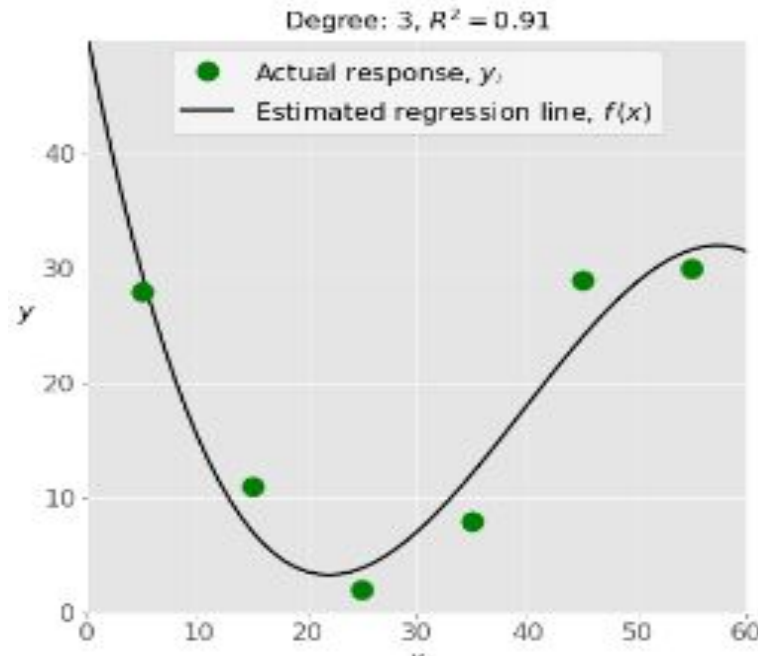
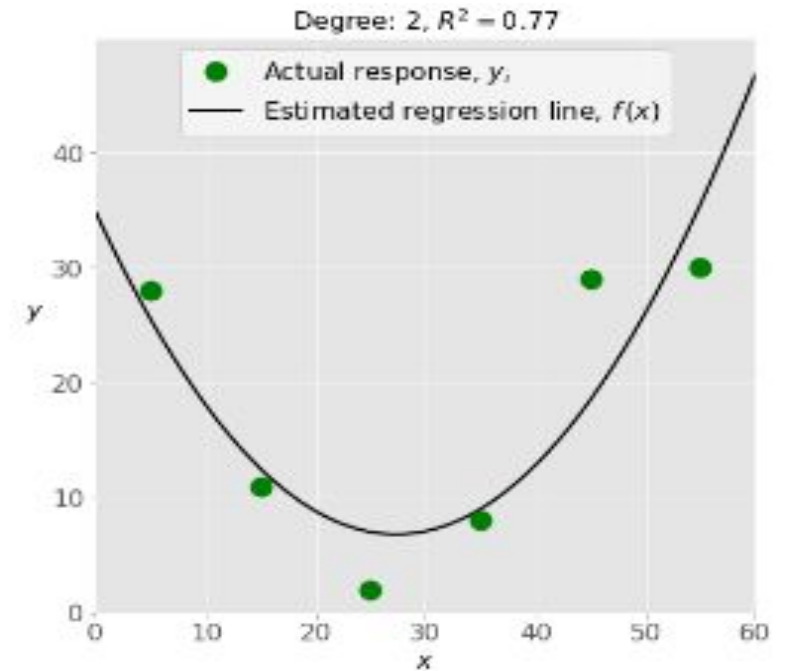
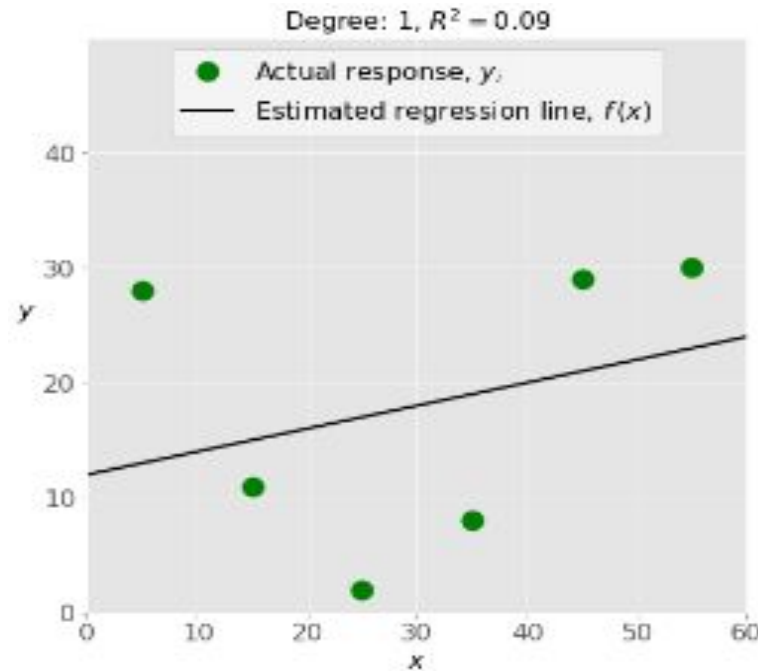
Examples of Polynomial Regression Models

<https://www.datasciencecentral.com/profiles/blogs/linear-regression-in-python-use-of-numpy-scipy-and-statsmodels>

Print equation of the curve in python

See here

<https://towardsdatascience.com/modeling-functions-78704936477a>



Python Codes for Curve Fitting: logarithmic; exponential; polynomial

- See examples here and try on your own:
- <https://www.kite.com/python/answers/how-to-do-exponential-and-logarithmic-curve-fitting-in-python>
- <https://heartbeat.fritz.ai/a-comprehensive-guide-to-logarithmic-regression-d619b202fc8>
- <https://towardsdatascience.com/modeling-functions-78704936477a>
- <https://towardsdatascience.com/machine-learning-polynomial-regression-with-python-5328e4e8a386>
- <https://towardsdatascience.com/introduction-to-linear-regression-and-polynomial-regression-f8adc96f31cb>
- <https://www.analyticsvidhya.com/blog/2020/03/polynomial-regression-python/>

Decision Tree

- See Week 3 Lectures and Lab Exercises

KNN (K-Nearest Neighbours) Regression

- We shall look at this in detail in KNN classification and regression

End